P-C-437
October 30, 1961

SUPERSEDING
Fed. Spec. P-S-751
September 15, 1955

FEDERAL SPECIFICATION

CLEANING COMPOUND, HIGH PRESSURE (STEAM) CLEANER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The alkaline steam cleaning compound covered by this specification is for use in steam cleaning machines for cleaning various ferrous and nonferrous surfaces.

1.2 Classification.

1.2.1 *Type and grade.* This specification covers one type and one grade only of steam cleaning compound.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

O-S-604—Sodium Metasilicate, Technical.
QQ-A-355—Aluminum Alloy, Plate and Sheet 2024.
QQ-A-561—Aluminum Alloy 1100 (2S); Plate and Sheet.
RR-S-366—Sieves, Standard for Testing Purposes.
PPP-D-705—Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 536—Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.
Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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Military Specifications:

MIL–L–15016—Lubricating Oil, General Purpose.

MIL–D–16791—Detergents, Nonionic.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The cleaning compound furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 6.3).

3.2 Material. The ingredient materials used in the manufacture of the steam cleaning compound covered by this specification shall be of high quality, intimately assembled and processed so as to produce a nonhygroscopic, nonsegregating, granular, free-flowing mixture in powder form which will show no evidence of caking in the as-received condition or during storage. When used in a steam cleaning machine of the high pressure-continuous tubular coil vapor generator type, the compound shall not impede nor clog the flow of solution by precipitation and building up of solids in the jets, coils, or orifices. The compound shall not adversely affect the user (skin burn or sneezing).

3.2.1 *Condition as received.* The steam cleaning compound in the as-received condition shall show no evidence of caking (for example, lumping or agglomerating) when tested as specified in 4.4.15.

3.3 Composition. The steam cleaning compound is not required to conform to definite chemical composition requirements. The manufacturer is given latitude in the selection of raw materials and processes of manufacture, provided the product meets all applicable requirements of this specification. The compound shall be free from fatty acid, rosin, soaps, starch, abrasives, gritty material, inert fillers, carbonates, bicarbonates, and free sodium hydroxide.

3.3.1 *Carbonates or bicarbonates.* The compound shall show no effervescence, when tested as specified in 4.4.1.

3.3.2 *Soap, rosin, and starch content.* The compound shall give no evidence of the presence of fatty acid, rosin, soap, or starch when tested in accordance with 4.4.4.

3.3.3 *Free sodium hydroxide.* When tested as specified in 4.4.11, not more than 0.2 ml. of 0.1 normal HCl shall be required to discharge the pink coloration with phenolphthalein of 15 ml. of the alcohol extract of the compound.

3.4 Performance requirements. The compound, steam cleaning, shall be equal to or superior in effectiveness to the comparison formula in 4.4.2.3 when tested as specified in 4.4.2.1 and 4.4.2.2. The compound covered by this specification shall be tested simultaneously with the standard comparison steam cleaning compound specified in 4.4.2.3.

3.4.1 *Water softening or stability.* When tested as specified in 4.4.2.1 at a temperature of 260°-265°F., a one-quarter percent solution of steam cleaning compound in water of 20-grain hardness (one-quarter gram of compound in 100 ml. of solution) shall give no evidence of precipitation or curd formation; nor shall there be an opalescence present greater than that produced in a solution of the standard comparison compound specified in 4.4.2.3 when tested simultaneously with the test steam cleaning compound covered by this specification. The opalescence shall be not greater than that given by the statistical upper limit set forth in 4.4.2.1.3.

3.4.2 *Cleaning efficiency.* The cleaning compound shall be equal to or superior to the standard comparison steam cleaning compound speci-

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fied in 4.4.2.3, prepared and tested under the same conditions, in ability to remove a standard soil when tested as specified in 4.4.2.2. The residual soil of any test compound determination shall be not greater than the statistical upper limit as set forth in 4.4.2.1.3.

3.5 Solubility. The insoluble matter in a 10-percent (10 gram of compound in 100 ml. of solution) distilled water solution of the compound shall not exceed 0.25 percent when tested at 25°C. as specified in 4.4.3.

3.6 pH value. The pH of a one-half percent solution of the cleaning compound (one-half gram of compound in 100 ml. of solution), in distilled water shall be not less than 10.5 nor more than 11.4, when tested at 25°C. as specified in 4.4.5.

3.7 Corrosion. Boiling solutions of the cleaning compound in distilled water shall cause no loss in weight of aluminum or aluminum alloy test specimens in excess of that indicated in table I when tested as specified in 4.4.6. There shall be no visible staining, discoloration, etching, or pitting of the test specimens.

TABLE I.—*Loss in weight permitted in corrosion test*

Percent solution	Fed. Spec. QQ–A–561 A1–1100	Fed. Spec. QQ–A–355 A1–2024
0.25	1.2 mg.	0.8 mg.
1.2	0.8 mg.	0.5 mg.

3.8 Surface tension. The surface tension of a one-quarter percent (one-quarter gram of compound in 100 ml. of solution) solution of the compound in distilled water at 25°C. shall be not more than 42 dynes per centimeter when determined as specified in 4.4.7.

3.9 Dust forming properties. The dust forming properties shall be such that, when tested as specified in 4.4.8, the dust shall settle within a period of five seconds as evidenced by no color change in red litmus paper.

3.10 Penetration. When tested as specified in 4.4.9, the average penetration shall be not less than 30.0.

3.11 Segregation. When tested as specified in 4.4.10, the difference of the percentages of any ingredient of the compound (such as silicates or total phosphates) taken from the different portions of a container shall not exceed 2 percent of the average percentage of that ingredient in the compound. The compound shall be subjected to the test specified in 4.4.10 at the discretion of the inspection laboratory when segregation appears to have occurred.

3.12 Fineness. The particle size of the compound shall be such that 100 percent shall pass a No. 6 mesh screen (U. S. Standard) when tested as specified in 4.4.12.

3.13 Rinsing. Solutions of the compound shall leave no white film on aluminum alloy when tested as specified in 4.4.13.

3.14 Caking in storage. When tested as specified in 4.4.14, the compound shall remain free flowing and of such a texture and granulation that 100 percent shall pass a No. 6 mesh, U. S. Standard screen.

3.15 Workmanship. The steam cleaning compound shall be manufactured in accordance with the best commercial practice to produce a high quality, stable, nonhygroscopic, nonsegregating, granular, free-flowing mixture in powder form.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the con-

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tract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 *Lot.* For purposes of sampling, a lot shall consist of material mixed or prepared at one time in one vessel. Material shall be identified by order of production or batch number.

4.2.2 *Sampling procedure.* From each inspection lot three separate 1-pound samples shall be taken. In case the material is produced by a batch process and the inspection lot contains more than 2 batches, the three samples shall normally be taken from different batches; from time to time, however, at the discretion of the inspector, two or three of the samples shall be taken from the same batch. In either case, samples shall be obtained in a manner calculated to disclose any nonuniformity of the material within the batch. Where material is produced by a continuous-run process, the three samples shall be taken so as to represent, respectively, the first part, the middle part, and the last part of the run which produced the inspection lot.

4.2.3 *Sampling for inspection of filled containers.* A random sample of filled containers shall be taken from each lot in accordance with Military Standard MIL–STD–105 at inspection level I, and acceptable quality level equals 2.5 percent defective, to verify compliance with all stipulations of this specification regarding fill, closure, marking and other requirements not involving tests.

4.2.4 *Inspection of filled containers.* Each sample filled container selected in accordance with 4.2.3 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected; and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected.

4.3 Tests. Testing under this specification shall be for the purpose of:

(*a*) Qualification.

(*b*) Acceptance of individual lots.

4.3.1 *Qualification tests (see 6.3).* Qualification tests shall consist of tests for all the requirements of this specification. For qualification testing a 10-pound sample is required plus a 125-pound sample in a metal drum as required for the caking-storage test. Interim approval will be granted a product passing all requirements of the specification except 3.14, subject to recission if 3.14 is failed.

4.3.2 *Acceptance tests.* The compound furnished for acceptance inspection shall be a product which has previously passed the qualification tests specified herein (see 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving qualification. Inspection tests for acceptance of individual lots shall normally consist of tests for water softening, soap and rosin contents, pH, carbonates and 1100 aluminum—¼ percent solution corrosion. The Government reserves the right to make any additional tests deemed necessary to determine compliance of the compound with all the requirements of this specification. If any sample is found to be not in compliance with this specification, the entire lot which it represents, if produced by continuous-run process, shall be rejected; or if the lot is produced in batches, all untested batches in the lot shall be rejected together with the batch from which the sample was taken.

4.4 Test methods.

4.4.1 *Carbonates or bicarbonates.* Approximately 6 normal hydrochloric acid shall be

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added drop-wise to a five percent (5 grams of compound in 100 ml. of solution) distilled water solution of the compound and examined for effervescence.

4.4.2 *Performance tests.* Performance tests shall consist of:

(*a*) Water softening or stability test.

(*b*) Cleaning efficiency.

4.4.2.1 *Water softening or stability tests.*

4.4.2.1.1 *Preparation of hard water stock solution.* A 40-grain per gallon hard water solution is prepared by dissolving 0.8090 g. of ACS grade calcium acetate, $Ca(C_2H_3O_2)_2 \cdot H_2O$, and 0.5546 g. of ACS grade magnesium sulphate, $MgSO_4 \cdot 7H_2O$, per liter of boiled distilled water.

4.4.2.1.2 *Water softening test.* To a test tube (200 mm. long by 25 mm. outside diameter) containing 20 ml. of a ½-percent distilled water solution of the compound (prepared by adding 500 milliliters of hot water (190°F.) to 10 grams of the compound in 2-liter volumetric flask, agitating to dissolve, cooling to room temperature, and diluting to volume with distilled water) add 20 ml. of 40-grain per gallon hard water stock solution, and mix well. Immerse in the solution, a test panel of WD 1020, 20-gage, cold-rolled steel, 3 inches by ½ inch in size, which has been polished with No. 1A3 emery cloth and cleaned with acetone. Place the test tube in a laboratory steam sterilizer at 260°-265°F. (steam pressure of 21-24 lb. per sq. in.) for 1 hour. Remove from the sterilizer. Cool to room temperature, and examine for precipitation; use an electrophotometer to measure opalescence in terms of optical density. Tests shall be run in quadruplicate. Prepare similar test solutions using the standard comparison compound of 4.4.2.3 and run in quadruplicate.

4.4.2.1.3 *Statistical upper limit.* The statistical upper limit at the .05 level in the water softening and cleaning tests of 4.4.2.1.2 and 4.4.2.2.4 may be determined with the following formula:

$$X + 0.72\,W$$

where: X = average of 4 determinations with the comparison compound.

W = difference between the maximum and minimum values in the four determinations with the comparison compound.

4.4.2.2 *Cleaning efficiency test.*

4.4.2.2.1 *Preparation of test panels.* Test panels of WD 1020, 20-gage, cold-rolled steel shall be prepared 2½ by 2½ inches in size. Each test panel shall have a hole ¼ inch in diameter placed ⅛ inch from one corner. Sharp edges shall be smoothed with No. 1A3 emery cloth. The panels shall then be degreased in ACS grade acetone. Both faces of each panel shall be polished with the same abrasive, stroking in one direction only. Clean in a hot alkaline solution until free from water-break. Rinse in water, then dip in absolute ethyl alcohol, and wipe dry with paper toweling.

4.4.2.2.2 *Standard soil.* The soiling material shall be Military symbol oil 3065 conforming to Military Specification MIL–L–15016. The cleaned panels, prepared as described in 4.4.2.2.1, are suspended on a "S" hook and dipped one at a time into a 400-ml. beaker of the oil maintained at a temperature of 26° ± 1°C. They are then removed and allowed to drain at a temperature of 26° ± 1°C. for 30 minutes. The remaining drop of oil in the lower corner of each panel is then removed with absorbent cotton. The amount of mineral oil adhering per panel shall be approximately 0.16 g.

4.4.2.2.3 *Preparation of cleaning solutions.* In a 2-liter beaker, place 1,600 ml. of ½ percent distilled water solution of the test compound (one-half gram of compound in 100 ml. of solution). A similarly prepared solution shall be made of the standard comparison compound. Bring solutions to a boil and keep at this temperature throughout the test. Maintain solution level throughout test by addition of distilled water as required.

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4.4.2.2.4 *Cleaning procedure.* Immerse the soiled test panels, prepared as specified in 4.4.2.2.2, in the cleaning solution by means of an iron or copper hook, one end of which passes through the ¼-inch hole and the other end over a glass rod placed across the top of the 2-liter beaker. At the end of 2½-minute and 5-minute intervals, respectively, the test panel is moved forward and backward, 3 times in each direction, and then up and down, 3 times in each direction; total agitation shall require not more than 6 seconds. At the end of 5 minutes immersion, the panel is removed from the cleaning solution and given 2 six-second rinses in distilled water (no agitation); there shall be a four-second drain between rinses. The rinsing solutions shall consist of 2 one-liter beakers, each containing 800 ml. of distilled water at 26° ± 1°C. The panel is then dried at 50°C. for 20 minutes, cooled and weighed. The panel is thoroughly washed with acetone, rinsed in absolute ethyl alcohol, dried with paper toweling and reweighed. The difference in weight is the amount of residual soil. Tests shall be run in quadruplicate.

4.4.2.3 *Standard comparison steam cleaning compound.* The composition of the standard comparison steam cleaning compound shall conform to table II.

TABLE II.—*Composition of standard comparison compound*

Component	Percent by weight
Sodium metasilicate, pentahydrate Fed. Spec. O–S–604	35.0
Primary sodium phosphate ACS grade $NaH_2PO_4 \cdot H_2O$ ...	10.5
Sodium tripolyphosphate technical grade $Na_5P_3O_{10}$	52.5
Nonionic surface-active agent MIL–D–16791, type 1	2.0

4.4.3 *Solubility.* Weigh 10 grams of the compound and add to 100 ml. of distilled water at 26° ± 1°C. in a 300-ml., tall-form beaker. Stir until compound is thoroughly dissolved. Let stand 10 minutes. Filter through a tared Gooch crucible with an asbestos mat. Wash with distilled water until washings give no alkaline reaction to phenolphthalein. Dry at 105°C. for 3 hours, cool and weigh. Gain in weight is insoluble matter.

4.4.4 *Soap, rosin, and starch content.*

4.4.4.1 *Soap, rosin.* Place 10 grams of the compound in a 100-ml. tall-form beaker. Add 30 ml. absolute ethyl alcohol. Place on steam plate and stir for 5 minutes. Filter through dry, double, No. 41 Whatman papers. Evaporate filtrate to dryness. Dissolve residue in 20 ml. distilled water. To half the solution add 5 cc. 1.0 *N* acetic acid. To the other half, add 5 cc. 1.0 *N* calcium chloride solution. Formation of precipitate in each solution indicates presence of soap, rosin or fatty acid.

4.4.4.2 *Starch.* Starch content shall be determined by method 6401, note 1, Federal Test Method Standard No. 536.

4.4.5 *pH value.*

4.4.5.1 *Apparatus.* The test shall be made with a pH electrometer having a sensitivity and readability of at least 0.05 pH. A sealed type, alkali-resistant glass electrode shall be used with a calomel reference electrode. The meter shall be standardized against a pH 10 buffer solution immediately before making a test. The instrument to be used shall be the Beckman pH Meter Laboratory Model G, or equal, and the electrode shall be the Beckman "Type E" high pH glass electrode or equal.

4.4.5.2 *Procedure.* Prepare 100 ml. of a ½ percent (one-half gram of compound in 100 ml. of solution) distilled water solution of the compound. Determine pH at 26° ± 1°C. No correction for sodium ion concentration shall be made.

4.4.6 *Corrosion test.*

4.4.6.1 *Preparation of test panels.* Test panels of 1100 aluminum shall be cut 3 inches by ¾ inch in size from approximately 0.034

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inch thick sheet aluminum conforming to Federal Specification QQ–A–561, H14 or H24 temper. Test panels of 2024-T3 aluminum shall be cut 3 inches by ¾ inch in size from approximately 0.025 inch thick sheet conforming to Federal Specification QQ–A–355. Sharp edges shall be smoothed with No. 4/0-150 alundum (Al_2O_3) polishing paper. Polish both faces with the same abrasive, stroking in one direction only. The 1100 aluminum panels are not to be re-used in this test. The 1100 aluminum panels should be lightly polished to avoid smearing of the surface, with approximately 18 milligrams of metal removed in polishing.

4.4.6.2 *Cleaning of panels.* Panels shall be cleaned with ACS grade acetone using a swab of absorbent cotton. They shall then be wiped with paper toweling, dipped in absolute ethyl alcohol, and again wiped with paper toweling.

4.4.6.3 *Corrosion test procedure.* Make up 0.25 and 1.2 percent solutions of the cleaning compound in distilled water (0.25 and 1.2 grams of compound in 100 ml. of solution). Tests shall be run in duplicate on each aluminum material in each solution. The cleaned test panel shall be weighed to 0.0001 gram and immersed completely in 200 ml. of the boiling solution of the compound in a 500-ml. tall-form beaker. The solution is kept at a boil throughout the test, the solution level being maintained by additions of distilled water as required. After 60 minutes, the test panel shall be removed from the boiling solution, rinsed under flowing tap water (cold), rinsed in distilled water (26° ± 1°C.), dipped in absolute ethyl alcohol, wiped dry with paper toweling, and reweighed. The presence of corrosion products shall be noted.

4.4.7 *Surface tension.* Prepare 100 ml. of ¼ percent (0.25 gram of compound in 100 ml. of solution) distilled water solution of the compound. Determine the surface tension at 26° ± 1°C., using a DuNouy tensiometer or interfacial tensiometer. Harkins-Jordan correction factors shall be applied.

4.4.8 *Dust forming.* The inside walls of a 250-ml. glass-stoppered graduate shall be rendered completely free of any grease and moisture. A 25-gram sample of the cleaner shall then be placed on the bottom of the clean graduate. The graduate shall then be stoppered, inverted and immediately returned to its original position. After any suspended dust has been allowed to settle for 5 seconds, the stopper of the graduate shall be removed and a moistened piece of red litmus paper suspended in the graduate without touching the walls so that the lower end of the litmus paper strip coincides with the 210 ml. mark at the upper end of the graduate. After 60 seconds, the litmus paper shall be removed and examined for any color change. The test shall be run in a constant humidity room, at a relative humidity of 50-55 percent and a temperature of 24° ± 1°C.

4.4.9 *Penetration.* Place 100 grams of the compound in a flat bottom, straight side, Pyrex glass crystallizing dish, 40 mm. high by 80 mm. diameter. Place sufficient lead shot into a 250-ml. low-form beaker so that the combined weight of the beaker and shot equals 800 grams. Place the weighted 250-ml. beaker on top of the 100 grams of compound. Place the assembly in a desiccator containing a saturated solution of potassium sulphate (K_2SO_4). Place the desiccator in an oven maintained at 43°-46°C. for 24 hours. Remove the assembly from oven and desiccator; remove weighted 250-ml. beaker from compound, and allow crystallizing dish containing compound to cool at room temperature for 2 hours. Determine penetration at three points on the surface, each at least 12 mm. from the side of the crystallizing dish, using the penetrometer and grease-penetrometer cone prescribed for method 311, Federal Test Method Standard No. 791, with a 5-second hold, 250 grams total weight of the cone and attachments and the temperature of the compound at 24° ± 1°C. Run test in duplicate.

4.4.10 *Segregation.* Samples taken from different portions of the container shall be analyzed for one constituent of the compound (such as total phosphates or silicates) in accordance with the applicable method of Federal Test Method Standard No. 536.

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4.4.11 *Free sodium hydroxide.* Place 10 grams of compound in a 100-ml. tall-form beaker. Add 30 ml. petroleum ether (B.P. 35-60°C.), and stir 3 minutes with a glass rod. Decant off petroleum ether. Dry residue in beaker until free from odor. Add 30 ml. absolute ethyl alcohol and stir 2 minutes. Decant off alcohol through dry, triple, No. 41 Whatman filters. Transfer 15 ml. of the filtrate to a 125-ml. flask and add several drops phenolphthalein indicator solution. If pink coloration results, titrate with 0.1 normal hydrochloric acid. Compare titration with requirements of 3.3.3.

4.4.12 *Fineness.* Transfer 100 grams of the compound to a No. 6 mesh screen conforming to Federal Specification RR–S–366, and place screen in a Ro-Tap sieve shaking apparatus. Sieve for 5 minutes. Determine amount retained on screen.

4.4.13 *Rinsing.* Test panels of 2024-T3 aluminum shall be cut 2½ by 2½ inches in size from approximately 0.025-inch-thick sheet conforming to Federal Specification QQ–A–355. They shall be polished and cleaned as specified in 4.4.6.1 and 4.4.6.2. Duplicate panels shall be tested. Immerse a panel for 5 minutes in 400 ml. of a boiling 1.0 percent solution of the cleaning compound in distilled water (1.0 gram of compound in 100 ml. of solution) contained in a 600-ml. beaker. Remove the panel, suspend at an angle of 45 degrees until dry. Then suspend the panel in about 1 liter of distilled water at 70° ± 2°C. for 3 minutes, withdrawing the panel slowly once each minute. Repeat the last step with a fresh sample of distilled water at 70°C. Remove the panel, and let dry at an angle of 45 degrees as before. When dry, examine the panel for residue. Then place a drop of absolute ethyl alcohol on the panel, allow to evaporate, and re-examine for white residue.

4.4.14 *Caking in storage.* The steam cleaning compound, 125 pounds net weight in a steel drum conforming to type III, 20-gage, of Federal Specification PPP–D–705, fitted with the specified lug cover, shall be stored in an unheated warehouse at Aberdeen Proving Ground, Maryland, for twelve months. The drum shall then be opened and the contents examined for free flowing and granular characteristics. If there is visual evidence of caking or lumping, transfer a single lump of the caked material obtained from the bottom third of the drum, and having a weight between 90 and 110 grams, to a No. 6 mesh, U. S. Standard screen conforming to Federal Specification RR–S–366. The sieve shall be placed in a Ro-Tap sieve shaking apparatus, and screened for 5 minutes. Examine the screen for 100 percent passage of material. Lumping of material that is broken up by the sieving operation to permit 100 percent passage thru the No. 6 mesh screen will not be cause for failure of this test.

4.4.15 *Condition as received.* An as-received container of the steam cleaning compound shall be emptied, and the contents examined for lumping or agglomerating of material. Visual evidence of either shall be cause for rejection of the compound.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging, levels A and C. Steam cleaning compound shall be packaged, 125 pounds net weight, in steel drums conforming to type III, of Federal Specification PPP–D–705, fitted with the specified lug cover.

5.2 Packing, levels A, B, and C. No overpacking is required.

5.3 Marking.

5.3.1 *Military agencies.* Containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

5.3.2 *Civil agencies.* Containers shall be marked in accordance with Federal Standard No. 123.

5.3.3 *Directions for use.* Unless otherwise specified, each container shall be durably and legibly marked with the directions for use of the compound as follows:

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Directions for use. The compound shall be thoroughly dissolved to give a concentration of approximately 10 ounces of the compound per gallon of water, using the soap tank of the cleaning machine, or a separate drum outside the cleaning machine and then transferring to the soap tank of the cleaning machine. The dry compound must be added to hot water in preparing the solution, and not vice versa. Adjust the concentrate solution metering device to obtain the desired results. A nozzle concentration of 0.3 percent (0.4 ounce per gallon) is suitable for average cleaning. When the machine is shut down at the end of a shift, a "blow-down" is essential for retarding scale formation in the coils of the machine.

5.3.4 In addition to the markings specified in 5.3, any special marking shall be as required by the order or contract.

6. NOTES

6.1 Intended use. The steam cleaning compound covered by this specification is intended for use in high pressure steam cleaning machines of the continuous tubular coil vapor-generating type, for cleaning of ferrous and nonferrous surfaces of equipment.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Unit quantity required—this compound should be purchased on the basis of net weight in pounds.

(*c*) Special marking (see 5.3.1 and 5.3.2).

(*d*) Administrative provisions for inspection records (see 4.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commanding General, Aberdeen Proving Ground, Maryland, Attention: Coating and Chemical Laboratory and information pertaining to qualification of products may be obtained from that activity.

6.4 Packaging. It has been found that compound packaged in drums permeable to moisture and carbon dioxide tends to cake in storage. Fiber drums especially are susceptible to this action.

6.5 Transportation description. Transportation description applicable to this item is:

Cleaning, scouring or washing compounds, not otherwise indexed by name, granular or powder.
Carload weight 36,000 pounds.

Cleaning, scouring or washing compounds, not otherwise indexed, granular or powder.
Truckload weight 36,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—O
Navy—Sh
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